Tutorial 5: Gas-Solid Fluidized Bed

Sajeev
CLL769: Applications of CFD

August 2025

1 Problem Statement & Objective

The objective is to perform Eulerian Two-Fluid Model (TFM) simulations of gas-solid fluidization in a pseudo-2D rectangular bed to investigate bubbling dynamics. The system involves a pseudo-2D domain ($0.2 \times 1.0 \times 0.02$ m). The primary phase is Air (gas), and the secondary phase is Glass beads (solid, $d_p = 2.26$ mm, $\rho_s = 2600$ kg/m^3). The simulation evaluates two inlet gas velocities: $U_g = 1.736$ m/s and $U_g = 2.78$ m/s.

2 Computational Setup

A velocity inlet for gas (0 m/s for solids) and a pressure outlet (0 Pa) were used. The walls were modeled with a no-slip condition for gas and the Johnson-Jackson condition for solids (specularity 0.05). The bed was initialized with a static bed height of 0.2 m patched with solid volume fraction $\alpha_s = 0.6$.

The Eulerian multiphase model (2 Phases, Phase-Coupled SIMPLE) was used. The drag was modelled using the Gidaspow model. Granular properties were handled via Kinetic Theory of Granular Flow (KTGF) using Syamlal-O'Brien (radial), Lun-et-al (pressure), and Schaeffer (friction). A 2nd Order Implicit transient scheme ($\Delta t = 0.001$ s) was used.

3 Geometry & Meshing

A uniform structured grid of 5,600 total cells was utilized (Resolution: 20 width, 70 height, and 4 depth). The pseudo-2D setup balances accuracy for capturing mesoscale structures with computational efficiency.

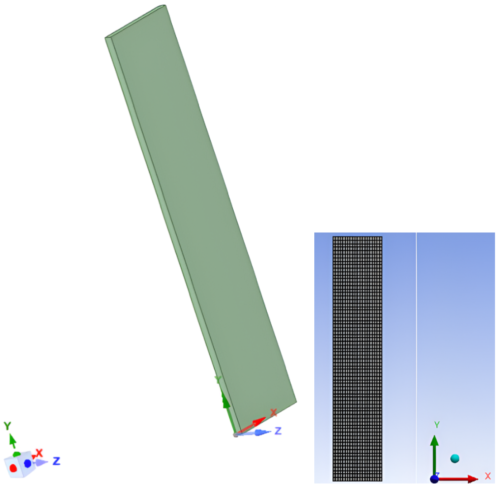

Figure 1: Domain geometry (left) and the uniform block-structured grid (right).

4 Results & Discussion

4.1 Fluidization Dynamics: Solid Volume Fraction (α_s)

At the higher superficial gas velocity (2.78 m/s), bubble formation is more intense. The bubbles are significantly larger, rise faster, and cause greater bed expansion and surface eruption compared to the lower velocity case (1.736 m/s).

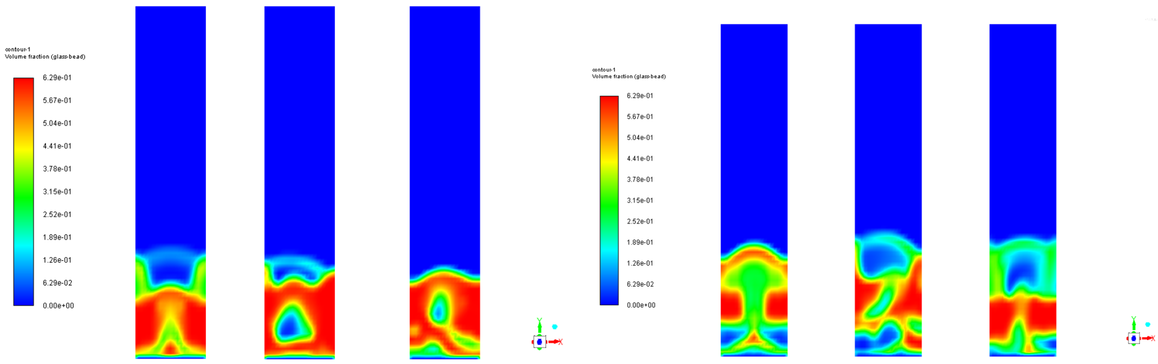

Figure 2: Solid Volume Fraction contours. Left: 1.736 m/s. Right: 2.78 m/s.

4.2 Fluidization Dynamics: Solid Velocity Magnitude

Solids are dragged rapidly upwards in the wake of the bubbles and recirculate downwards near the walls. The higher gas velocity substantially increases the peak solid velocities and overall bed mixing.

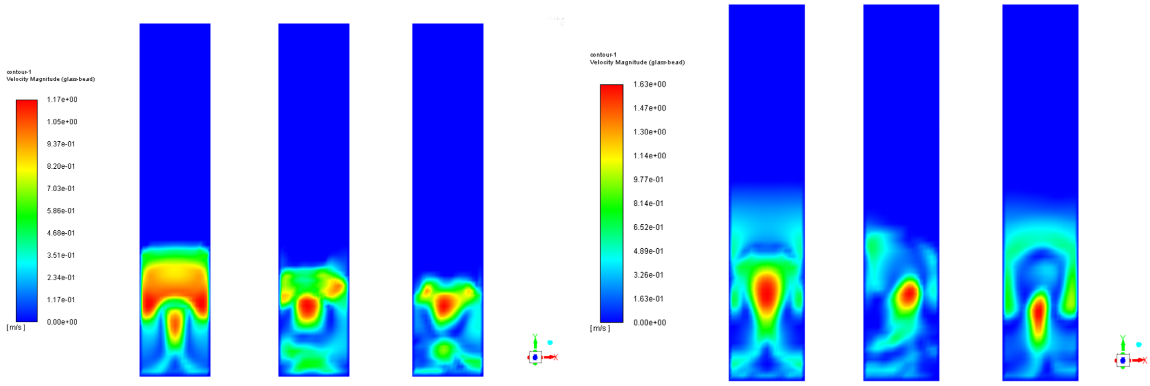

Figure 3: Solid Velocity Magnitude contours. Left: 1.736 m/s. Right: 2.78 m/s.

4.3 Time Evolution at Monitored Probe

Analyzing the transient probe data at a fixed point ($x = 0.1, z = 0.01$ m), sharp drops in α_s corresponding to spikes in solid velocity indicate the passing of a bubble through the probe location. The higher velocity case exhibits a much higher frequency and intensity of bubbling.

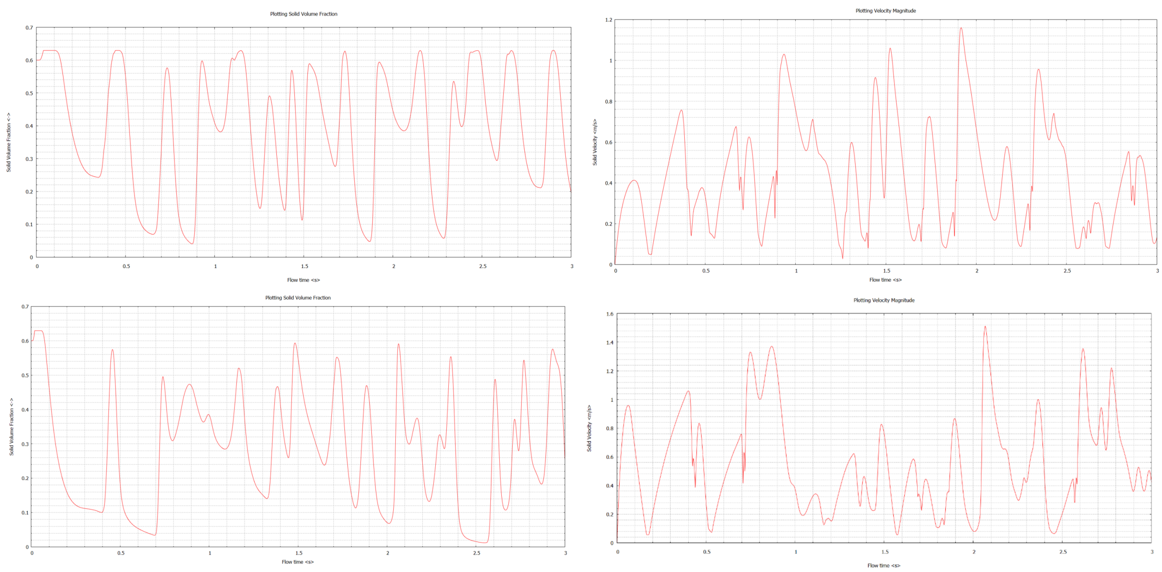

Figure 4: Probe data for solid volume fraction and solid velocity. Top row: 1.736 m/s. Bottom row: 2.78 m/s.